

Bounded weak solutions have exactly bounded VMO traces

Same-paper consequence clarifying the VMO bullet of arXiv:2605.00439

11 August 2026

Status. Verified literal classification implied by the source itself, not claimed as a new solution of the intended broader VMO program. Under the source’s bounded-solution definition, the admissible VMO traces are exactly $\text{VMO} \cap L^\infty$; hence no extension to genuinely unbounded VMO data is possible without changing the solution class.

1 Source problem and answer

Auscher and Bechtel study

$$\partial_t u - \text{div}(a(t, x, u) \nabla u) = 0, \quad u(0) = u_0, \quad (1)$$

on $\mathbb{R}^n$. Their Definition 1.2 calls u a *bounded weak solution* only if $u \in L^\infty((0, T) \times \mathbb{R}^n)$ and

$$\langle u(t), \psi \rangle \longrightarrow \langle u_0, \psi \rangle \quad (t \downarrow 0), \quad \psi \in C_c^\infty(\mathbb{R}^n). \quad (2)$$

Corollary 1.5 proves existence for every $u_0 \in L^\infty$ satisfying the source range condition. The paper then asks whether this can be extended to “more irregular data such as $u_0 \in \text{VMO}(\mathbb{R}^n)$ ” [1].

Open problems. We conclude with some open problems:

- (Systems) Our Proposition 4.1 remains valid for systems, hence it would be interesting to look at quasilinear systems with our approach.
- (Uniqueness) For irregular initial data such as $u_0 \in L^\infty(\mathbb{R}^n)$, it would be interesting to investigate suitable conditions ensuring uniqueness of bounded weak solutions. The issue is to have sufficient a priori regularity near the initial time.
- (Existence) It would be interesting to investigate whether the existence result in Corollary 1.5 can be extended to more irregular data such as $u_0 \in \text{VMO}(\mathbb{R}^n)$.

The answer is no for the stated solution class. In fact there is an exact classification.

Theorem 1 (Classification of VMO traces). *Let a and the open set O satisfy the hypotheses of [1, Assumption A]. Let $u_0 \in \text{VMO}(\mathbb{R}^n)$ be a locally bounded representative whose essential range is contained in O . There exists a bounded weak solution in the sense of [1, Definition 1.2] only if $u_0 \in L^\infty(\mathbb{R}^n)$. Conversely, if $u_0 \in L^\infty(\mathbb{R}^n)$ and its closed essential range is contained in O , such a global solution exists. Thus the source’s exact VMO existence class is*

$$\text{VMO}(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$$

(with the same range condition as in the source).

The converse is precisely [1, Corollary 1.5]. The necessity is already contained abstractly in [1, Proposition 3.4(b)]; the next PDE-free trace lemma gives a one-line independent proof. To show that the boundary is nonvacuous under every standard global VMO convention, Section 3 constructs a locally bounded but unbounded datum lying even in CMO, the closure of C_c^∞ in BMO.

2 The universal bounded-trace obstruction

Lemma 2 (Weak-* closedness of the L^∞ ball). *Suppose $u \in L^\infty((0, T) \times \mathbb{R}^n)$ and a distribution f satisfies*

$$\int_{\mathbb{R}^n} u(t, x) \psi(x) dx \longrightarrow \langle f, \psi \rangle \quad (t \downarrow 0)$$

for every $\psi \in C_c^\infty(\mathbb{R}^n)$. Then f is represented by a function in $L^\infty(\mathbb{R}^n)$ and

$$\|f\|_\infty \leq \|u\|_{L^\infty((0, T) \times \mathbb{R}^n)}.$$

Proof. Put $M = \|u\|_\infty$. By Fubini, there is a full-measure set $E \subset (0, T)$ such that $\|u(t, \cdot)\|_\infty \leq M$ for every $t \in E$. Choose $t_j \in E$ with $t_j \downarrow 0$. Then

$$|\langle f, \psi \rangle| = \lim_j \left| \int u(t_j, x) \psi(x) dx \right| \leq M \|\psi\|_1.$$

Thus f extends by density to a bounded functional of norm at most M on $L^1(\mathbb{R}^n)$. The duality $(L^1)^* = L^\infty$ represents that functional by some $g \in L^\infty$ with $\|g\|_\infty \leq M$. Hence $f = g$ as a distribution. If the prescribed datum was already a locally integrable function, it equals g almost everywhere. $\square$

Applying Lemma 2 to (2) proves the necessary half of Theorem 1. Notice that ellipticity, regularity of a , and even equation (1) play no role: the obstruction is built into the simultaneous requirements of uniform spacetime boundedness and distributional attainment of the initial datum.

3 A locally bounded unbounded CMO datum

There are two widespread conventions on $\mathbb{R}^n$: Sarason VMO, the BMO-closure of uniformly continuous BMO functions, and the smaller CMO, the BMO-closure of C_c^∞ . The following example lies in the smaller space.

Lemma 3. *For every $n \geq 1$, with $A = e^e$, the radial function*

$$f(x) = \log \log(A + |x|) \tag{3}$$

is locally bounded, unbounded, and belongs to CMO($\mathbb{R}^n$).

Proof. The first two claims are immediate. We verify Uchiyama's three mean-oscillation conditions, which characterize the BMO-closure of C_c^∞ [2]. Balls and cubes give equivalent conditions. For a ball B , write

$$\text{osc}_B(f) = \frac{1}{|B|} \int_B |f - f_B|.$$

First, f is globally Lipschitz: away from the origin,

$$|\nabla f(x)| = \frac{1}{(A + |x|) \log(A + |x|)} \leq \frac{1}{Ae},$$

and the same bound follows across the origin from the radial profile. Consequently

$$\sup_{\text{rad}(B) \leq r} \text{osc}_B(f) \leq \frac{2r}{Ae} \rightarrow 0 \quad (r \downarrow 0). \quad (4)$$

We next estimate centered balls. Put $h(r) = \log \log(A + r)$ and $L_R = \log(A + R)$. In radial coordinates $s = |x|/R = e^{-y}$, the normalized measure is $ne^{-ny} dy$. Since $A + Re^{-y} \geq e^{-y}(A + R)$, for $0 \leq y \leq L_R/2$,

$$0 \leq h(R) - h(Re^{-y}) \leq -\log(1 - y/L_R) \leq \frac{2y}{L_R}.$$

For $y > L_R/2$, use the crude bound $h(R) - h(Re^{-y}) \leq \log(L_R/e)$. It follows that

$$\int_{B(0,R)} |f - h(R)| \leq \frac{2}{nL_R} + \log(L_R/e)e^{-nL_R/2} \rightarrow 0. \quad (5)$$

Mean oscillation is at most twice the average distance from any constant.

Now let $B = B(c, r)$ with $r \rightarrow \infty$. If $|c| \leq 2r$, then $B \subset B(0, 3r)$ and comparison with the constant $h(3r)$, followed by (5), gives

$$\text{osc}_B(f) \leq 2 \int_{B(0,3r)} |f - h(3r)| \rightarrow 0.$$

If $|c| > 2r$, the gradient bound on B instead gives

$$\text{osc}_B(f) \leq \frac{4r}{|c| \log(A + |c|/2)} \leq \frac{2}{\log(A + r)} \rightarrow 0.$$

This proves uniform vanishing on large balls. Finally, for any fixed ball translated to infinity, the same gradient estimate makes its oscillation tend to zero. Together with (4), these are Uchiyama's three conditions. They also imply $f \in \text{BMO}$ (the remaining bounded range of radii and centers is compact, while the far-center gradient bound is uniform). Therefore $f \in \text{CMO}(\mathbb{R}^n)$. $\square$

Corollary 4 (Explicit counterexample). *Take $O = \mathbb{R}$ and $a(t, x, y) = I_n$. Then (1) is the heat equation and satisfies every source assumption. For the locally bounded datum (3), there is no bounded weak solution in the source sense.*

Proof. Lemma 3 gives $f \in \text{CMO} \subset \text{VMO}$ and $f \notin L^\infty$. Lemma 2 rules out the required bounded solution before the heat equation is used. $\square$

4 Scope and interpretation

The result does not say that VMO data cannot seed a useful parabolic theory. It says that genuinely unbounded VMO data cannot do so while the solution remains essentially bounded on the whole cylinder down to $t = 0$ and attains the datum distributionally. A meaningful extension must therefore change at least one ingredient, for example by allowing a time-dependent bound that diverges at the initial time or by using a BMO/VMO-adapted solution norm. If “VMO data” was intended to mean only $\text{VMO} \cap L^\infty$, existence is already contained in Corollary 1.5 of the source.

The current v3 source already proves that every bounded linear solution has an L^∞ trace in Proposition 3.4(b), and every bounded quasilinear solution is such a linear solution after freezing its coefficient. Thus this note is a same-paper logical consequence, not a novelty claim. Its purpose is to identify precisely how any nontrivial VMO reformulation must differ.

References

- [1] P. Auscher and S. Bechtel, *Existence and uniqueness of weak solutions to quasilinear PDEs with critical data*, arXiv:2605.00439v3, 2026.
- [2] A. Uchiyama, *On the compactness of operators of Hankel type*, Tohoku Math. J. (2) **30** (1978), 163–171; see the CMO characterization on p. 166.